

Marcos Javier Alvarez

Senior Full-Stack Engineer · LLM and agentic systems in production · TypeScript, Node.js, React, AWS

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Portfolio, references and a queryable MCP endpoint: hire-me-mcp-web.vercel.app · **Query it from any MCP client:**

<https://hire-me-mcp-web.vercel.app/api/mcp>

Summary

Senior full-stack engineer (TypeScript, Node.js, React, AWS) who has spent the last three years putting LLM and agentic features into production on a mortgage-lending platform: evaluation harnesses with human-graded ground truth and LLM-as-judge scoring, per-call cost tracing, version-controlled prompts and provider failover. I built and maintain the platform's communications service (57,000+ emails, SMS and calls triaged in a recent seven-month window, ~70% reaching an automated triage outcome) and led the migration off a prompt-management vendor to repository-native LLM infrastructure across four services with no significant regression. Thirteen years of full-stack, DevOps and client-facing delivery underneath, plus two open-source AI projects (an MCP server and an agent-orchestration CLI).

Experience

House Numbers — Senior Full Stack Engineer · May 2022 – Present

- Built and maintain the communications service that triaged 57,000+ emails, SMS and calls in a recent seven-month window, ~70% to an automated outcome, letting operations retire its manual-triage shifts: webhook intake, LLM-assisted extraction and classification kept separate from deterministic routing, loan matching over field-level-encrypted data, hardened with alerts, durable retries and webhook reconciliation.
- Retired a prompt-management vendor (Orq.ai) and its recurring cost by leading a multi-month, service-by-service migration to repository-native LLM infrastructure across four services with no significant regression: version-controlled prompts, provider failover, shared per-call cost tracing and automated evaluations that let the team swap in cheaper models under measured quality.
- Replaced hallucination-prone LLM-generated loan-document checks with deterministic, tested validation (plus capped, evaluated AI observations) in under two weeks, after decomposing the monolithic analysis pipeline into three idempotent, retryable stages; processors adopted the checks and the open-ended LLM runtime cost for them was removed.
- Eliminated failed-upload complaints on a public document-upload flow where an estimated two in three submissions had ended in a complaint or escalation: designed and built the replacement boundary (opaque sessions, CAPTCHA, rate limiting, replay protection, encrypted audit trail) and a hybrid deterministic/LLM routing model that kept ambiguous matches with a human.
- Owned incident response on the document-processing path: contained a crash loop (96 restarts in 24 hours against a 0-4 baseline), made the failing path observable and retryable, and reprocessed the affected window with no permanent data loss.
- Authored the shared on-call investigation workflow the whole rotation adopted, and proposed the alert-to-PR automation built on top of it: today a human report or a New Relic alert can trigger an agent that reviews the available evidence and develops likely causes, and prepares a pull request when the issue maps to a code change, for an engineer to validate and merge.

Tech: TypeScript, Node.js, React, MongoDB, AWS ECS, Redis, OpenAI API, Anthropic API, New Relic, Zod.

Xogito Group — Senior Software Development Engineer · Dec 2020 – Mar 2022

- Main backend engineer (NestJS, TypeScript, PostgreSQL, AWS, GitLab CI) on a multi-tenant SaaS for real-time quality-control inspections; owned backend architecture and delivery end to end.
- Recovered a frustrated client's account after the project manager resigned: became the team's spokesperson, paired quick wins with core-workflow repairs, rebuilt trust within a few sprints and led the client product conversations for about six months; the client went on to commission additional projects.
- Onboarded and mentored new engineers with an approach later adopted at House Numbers as a rotating buddy program; 15+ people onboarded across the two companies.
- Led a legacy-SAP data migration more complex than the FullStack Labs one below: evaluated ETL tooling, built custom scripts where existing tools could not represent the models, validated rounding compatibility early, and needed no post-migration reconciliation.

Tech: NestJS, TypeScript, PostgreSQL, Docker, GitLab CI, AWS.

FullStack Labs — Senior Software Engineer · Dec 2018 – Dec 2020

- Owned the migration of historical quote data out of a legacy SAP system with zero data loss: custom ETL, traced a financial-total discrepancy to SAP/JavaScript rounding differences, and ran a frozen cutover with snapshot-backed reconciliation; users retired the legacy system after MVP launch.

Tech: Node.js, React, PostgreSQL, AWS, Docker.

Earlier Experience

Belatrix Software — Senior NodeJS Software Developer · Jul 2018 – Nov 2018

- Acting technical lead: serverless architecture (Lambda, CloudFormation, DynamoDB) for a dynamic-document platform; led a four-person team delivering it.

Rokk3r — Senior Full Stack Developer and DevOps · Feb 2016 – Jun 2018

- Via Kubesoft SAS: migrated a real-time telecom monolith to an AWS Kubernetes microservices cluster; migrated a mobile backend off Parse ahead of its shutdown deadline.

Kubesoft SAS — Senior Full Stack & Mobile Developer · Jan 2015 – Jun 2018

- Web/hybrid-mobile products for Colombian and international clients; built the first backend of Mutual, a hackathon-winning parenting app.

Jarvi Games — Full Stack Developer and DevOps · Feb 2013 – Jun 2015

- Real-time game backend with micropayments (Node.js, MongoDB, PostgreSQL, PHP) and its infrastructure.

Selected Projects

hire-me-mcp — Creator and maintainer · github.com/garusis/hire-me-mcp · hire-me-mcp-web.vercel.app

- Public MCP server, RAG-backed interview agent (Neon pgvector, Gemini embeddings) and Next.js site over one typed career dataset; Vitest, Playwright e2e against real Vercel previews, and retrieval/agent evals as required CI gates.

cowork — Creator and maintainer · github.com/garusis/cowork

- Open-source Python CLI that orchestrates multiple AI coding agents into role-based teams, with phase gates, per-role evals and cost reporting.

Skills

- **AI / LLM Engineering:** OpenAI API, Anthropic API, LLM evaluation (human-graded, LLM-as-judge), AI Agents, Prompt Engineering, Document AI (OCR/vision-LLM), MCP (Model Context Protocol), RAG (Retrieval-Augmented Generation).
- **Languages:** TypeScript, JavaScript, Python, SQL.
- **Frontend:** React, Next.js.
- **Backend:** Node.js, NestJS, Microservices, Event-Driven Architecture, REST APIs, Webhooks.
- **Data:** MongoDB, PostgreSQL, Redis, pgvector.
- **Cloud & Infrastructure:** AWS, Docker, Kubernetes, Serverless (Lambda/CloudFormation), GitLab CI, GitHub Actions, Vercel, New Relic, AWS ECS.
- **Practices:** CI/CD, Observability, Turborepo/pnpm monorepos, Mentoring & Onboarding, PII handling and field-level encryption, Playwright, Vitest, Zod, ETL / Data Migration.

Education

- Universidad Nacional Abierta y a Distancia (UNAD) — coursework toward B.S. Systems Engineering